

SD-WAN Configuration Documentation

HQ and Branch FortiGate Deployment

1. Network Topology Overview

The network consists of two sites:

Headquarters (HQ)

WAN Path

- Two routers provide dual WAN paths between sites:
 - **R1**: 10.10.10.1
 - **R2**: 10.20.20.1
- Both HQ and Branch FortiGate units connect to an IOU1 switch via VLAN trunk links.

2. HQ FortiGate Configuration

2.1 WAN Interface Sub-Interfaces

Two VLAN-based WAN interfaces were created for the dual routers.

GUI Path:

Network → Interfaces → Create New → Interface

Parameter	WAN_R1	WAN_R2
Type	VLAN	VLAN
Parent Interface	port5	port5
VLAN ID	10	20

IP	10.10.10.2/30	10.20.20.2/30
Role	WAN	WAN
Access	PING	PING

CLI Equivalent

```

config system interface
  edit WAN_R1
    set vdom root
    set ip 10.10.10.2 255.255.255.252
    set allowaccess ping
    set role wan
    set interface port5
    set vlanid 10
  next
  edit WAN_R2
    set vdom root
    set ip 10.20.20.2 255.255.255.252
    set allowaccess ping
    set role wan
    set interface port5
    set vlanid 20
  next
end

```

Connectivity Test

```

execute ping 10.10.10.1 → Success
execute ping 10.20.20.1 → Success

```

2.2 SD-WAN Zone and Members

GUI Path:

Network → SD-WAN → SD-WAN Zones

Zone Name: HQ_SD-WAN

Members Added:

Member	Interfac	Gateway
1	WAN_R1	10.10.10.1
2	WAN_R2	10.20.20.1

2.3 Performance SLAs

GUI Path:

Network → SD-WAN → Performance SLAs

SLA Name	Monitored Interface	Probe Target	Metrics
SLA_R1	WAN_R1	8.8.8.8	Latency, Jitter, Packet Loss
SLA_R2	WAN_R2	1.1.1.1	Latency, Jitter, Packet Loss

Thresholds:

- Latency: **50 ms**
- Jitter: **20 ms**
- Packet Loss: **5%**

SLA status verified as UP.

2.4 SD-WAN Rule

GUI Path:

Network → SD-WAN → SD-WAN Rules

Field	Value
Rule Name	HQ-BR-VPN
Source	all
Destination	Branch remote subnets (HQ-To-BR objects)
Strategy	Best Quality

Preferred Interfaces	WAN_R1, WAN_R2
Measured SLA	SLA_R1
Quality Criteria	Latency

3. Branch FortiGate Configuration

Branch configuration mirrors HQ, using _BR suffixes.

3.1 WAN Sub-Interfaces

Parameter	WAN_R1_BR	WAN_R2_BR
Type	VLAN	VLAN
Parent Interface	port6	port6
VLAN ID	10	20
IP	10.10.10.x/30	10.20.20.x/30
Role	WAN	WAN
Access	PING	PING

CLI

```

config system interface
    edit WAN_R1_BR
        set ip 10.10.10.x 255.255.255.252
        set vlanid 10
        set allowaccess ping
        set interface port6
    next
    edit WAN_R2_BR
        set ip 10.20.20.x 255.255.255.252
        set vlanid 20
        set allowaccess ping
        set interface port6
    next
end

```

3.2 SD-WAN Zone

Zone Name: BR_SD-WAN

Members:

Interface	Gateway
WAN_R1_BR	10.10.10.1
WAN_R2_BR	10.20.20.1

3.3 Performance SLAs

SLA Name	Interface	Target
SLA_BR_R1	WAN_R1_BR	10.10.10.1
SLA_BR_R2	WAN_R2_BR	10.20.20.1

Thresholds identical to HQ.

3.4 SD-WAN Rule

Field	Value
Name	BR-TO-HQ-VPN
Source	all
Destination	HQ remote subnets
Strategy	Best Quality
Preferred Interfaces	WAN_R1_BR, WAN_R2_BR
Measured SLA	SLA_BR_R1
Criteria	Latency

4. Status Summary

Component	HQ	Branch
WAN Interfaces	OK	OK

Gateway	OK	OK
Reachability		
SD-WAN Zone	Configured	Configured
SLA Monitoring	Active	Active
Rules	Applied	Applied